

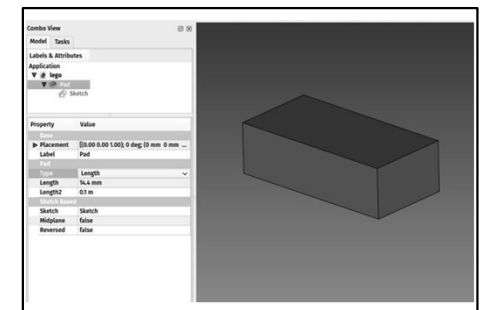
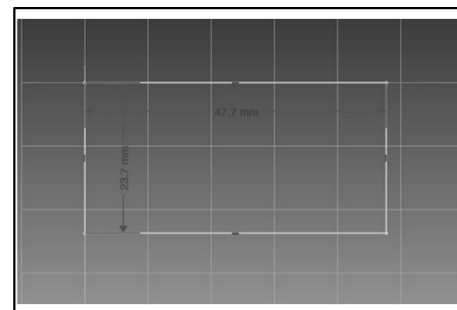
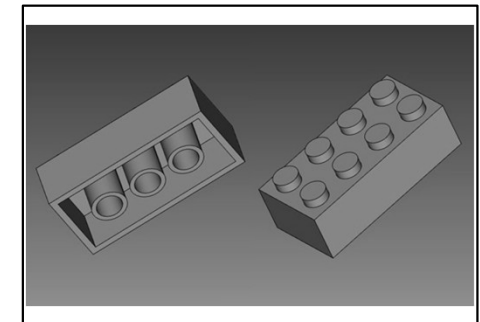
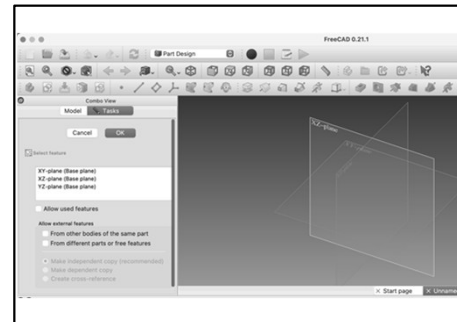
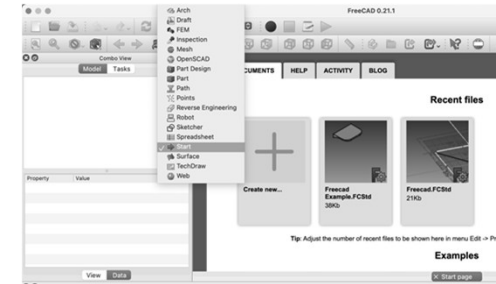
Module: 3D modeling using software

Lecture 7 · Module: 3D modeling using software Guided practice with FreeCAD 1.0

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Faculty of Engineering and Technology



The Software of 3D Modeling for 3D Printing



FreeCAD interface, workbench selection, constrained sketch, pad, and final tutorial part

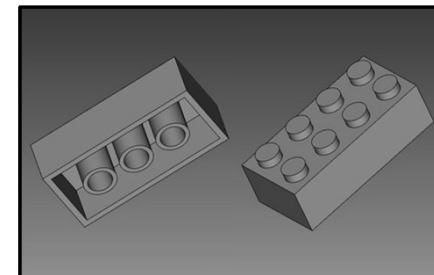
What students should achieve today

Learning outcomes

- Explain what a CAD model stores beyond the final rendered view.
- Relate FreeCAD concepts to OpenGL ideas: geometry, transforms, viewing, and display.
- Build a simple 3D part through a guided workflow in class.
- Describe the software pipeline: document → body → sketch → constraints → 3D features → revision.

Lecture flow

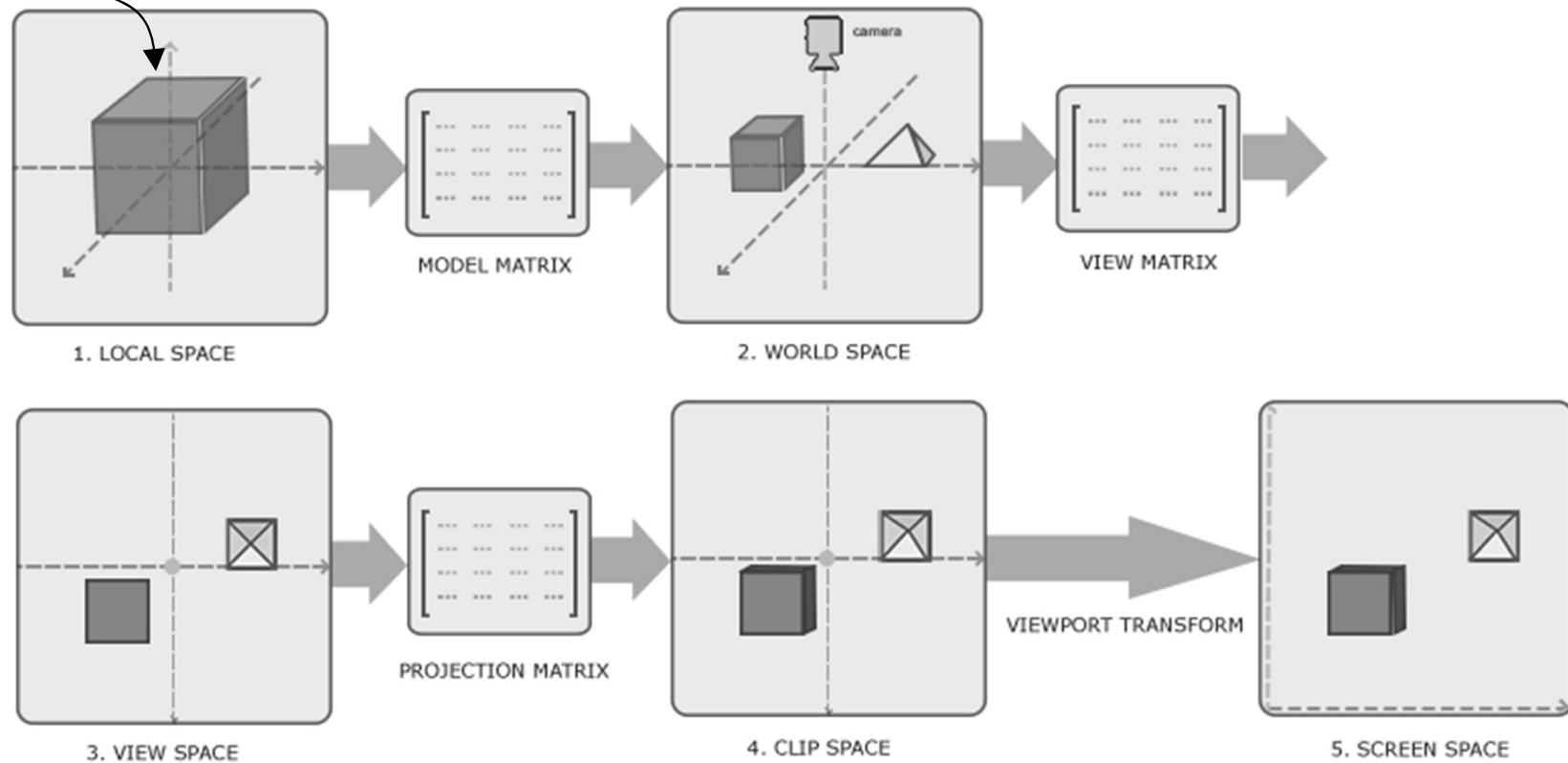
1. Concept bridge: OpenGL thinking → CAD thinking
2. Interface and workbench orientation
3. Parametric modeling pipeline
4. Guided build: sketch, pad, pocket, top features
5. Independent mini-challenge and recap



*Tutorial end state used as the classroom
reference model*

Coordinate Spaces in OpenGL

3D modeling
software to create
geometry

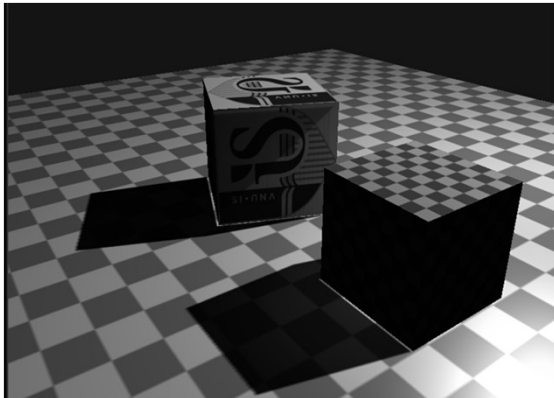


OpenGL to CAD: what changes? (1/2)

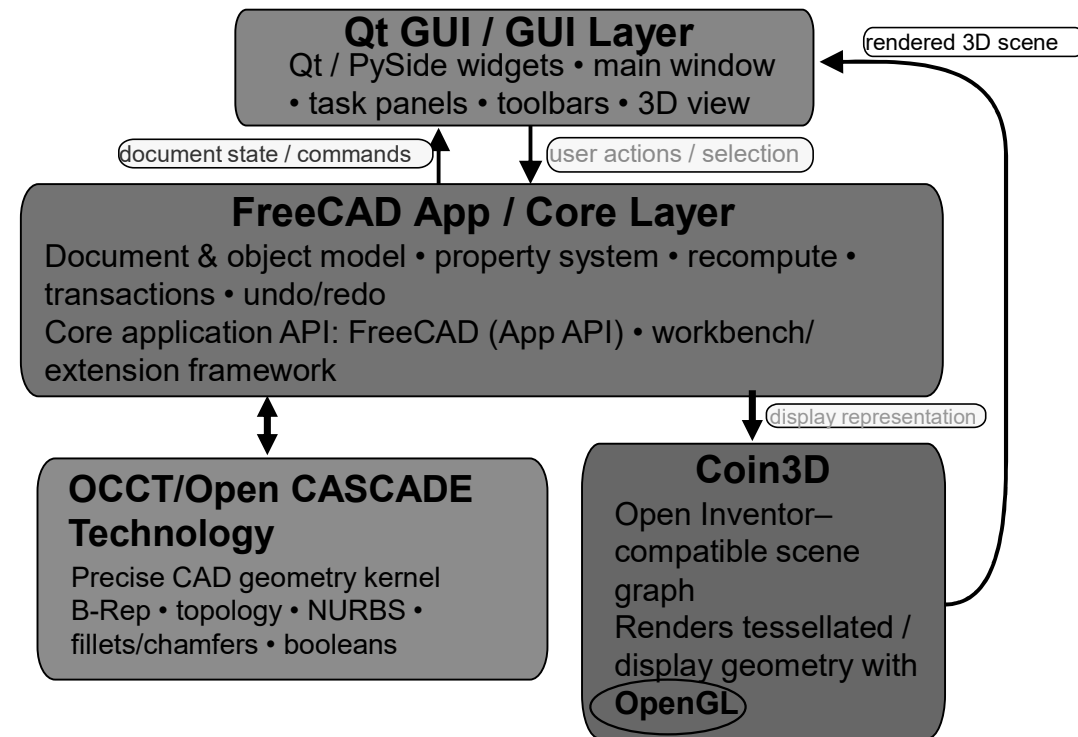
What students already know in OpenGL

- Objects appear through vertices, transforms, materials, lights, and a camera/view pipeline.
- A rendered scene is optimized for display.
- The programmer usually decides how geometry is represented and updated.

OpenGL is a rendering API



FREECAD ARCHITECTURE

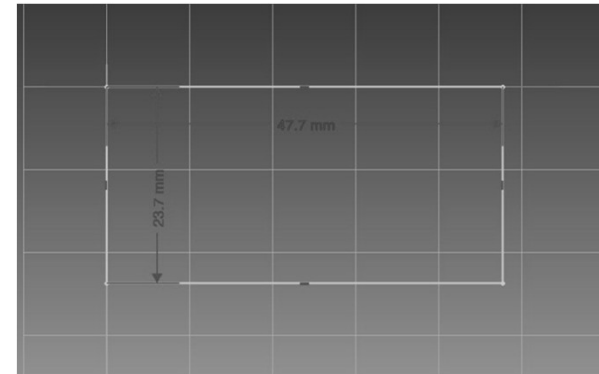


— geometry operations — display / tessellation
— user events — API / workbench access

OpenGL to CAD: why the pipeline matters (2/2)

A useful classroom analogy

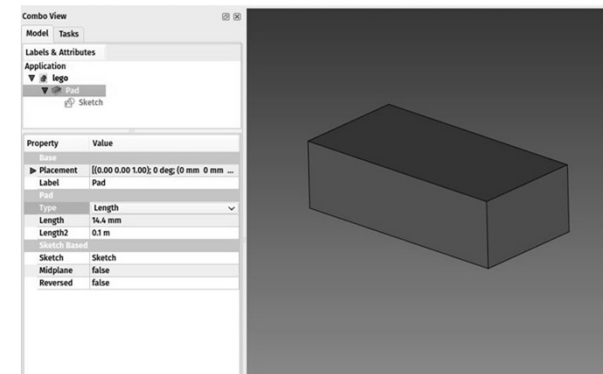
- **OpenGL** asks: “How do I draw this object now?”
- **CAD** asks: “How was this object built, and what should happen if one dimension changes?”
- In CAD, a hole diameter is not just geometry; it is a named, editable design decision.



Fully constrained sketch

Key consequence for engineering

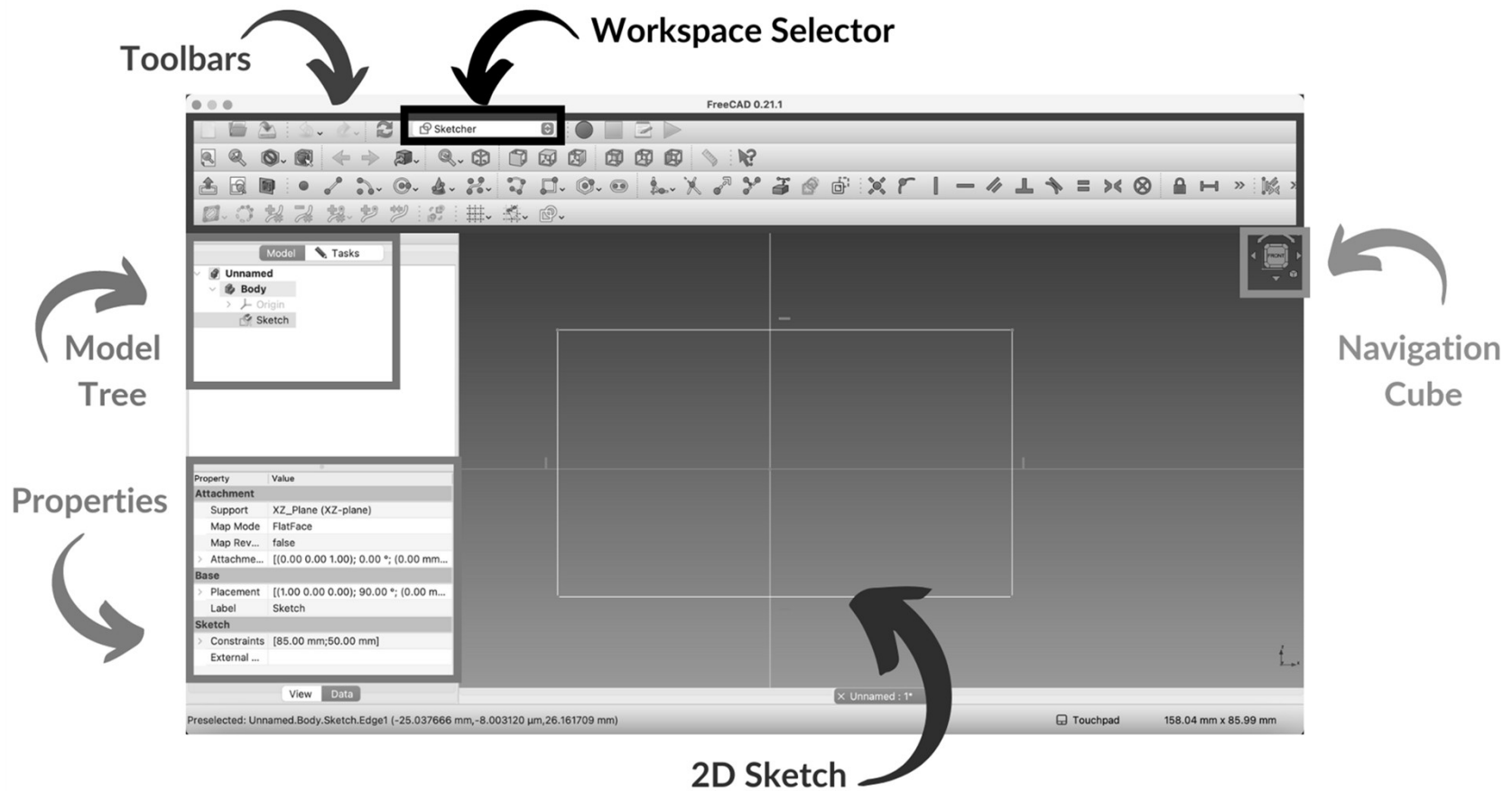
- Stable edits are more valuable than one perfect screenshot.
- A good model survives revision without starting over.
- That is why we will focus on constraints, pad, pocket, and feature order.



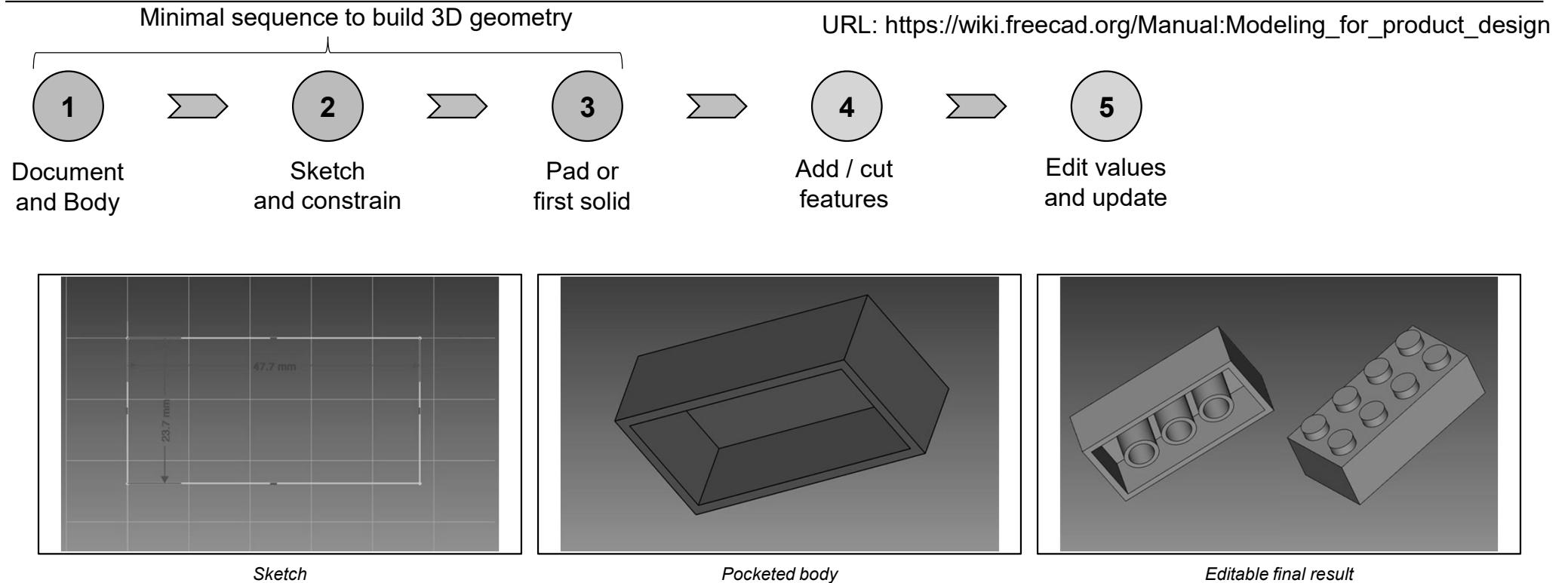
The next feature depends on that sketch

FreeCAD interface and workbench orientation

FreeCAD GUI

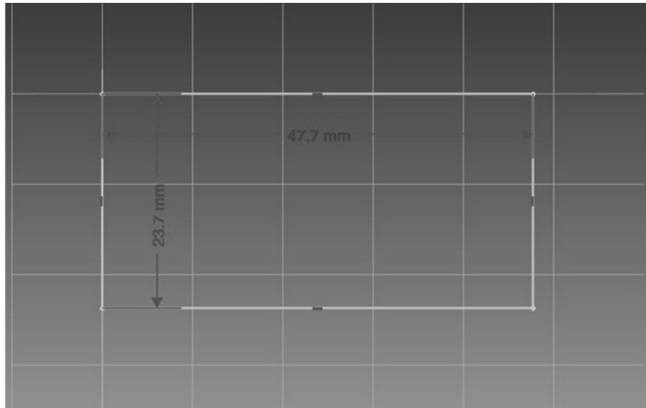


The modeling pipeline in software

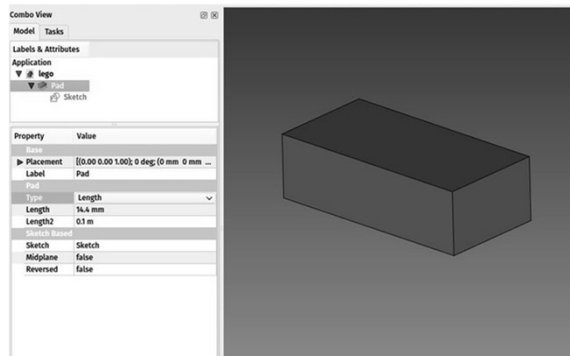


- A feature-based model is valuable because any later step still depends on earlier design choices.
- When the model tree is healthy, changing a dimension should regenerate the part instead of forcing a redraw.

Sketcher + Part Design: the two ideas to remember



Sketcher: geometry + constraints → predictable 2D profile

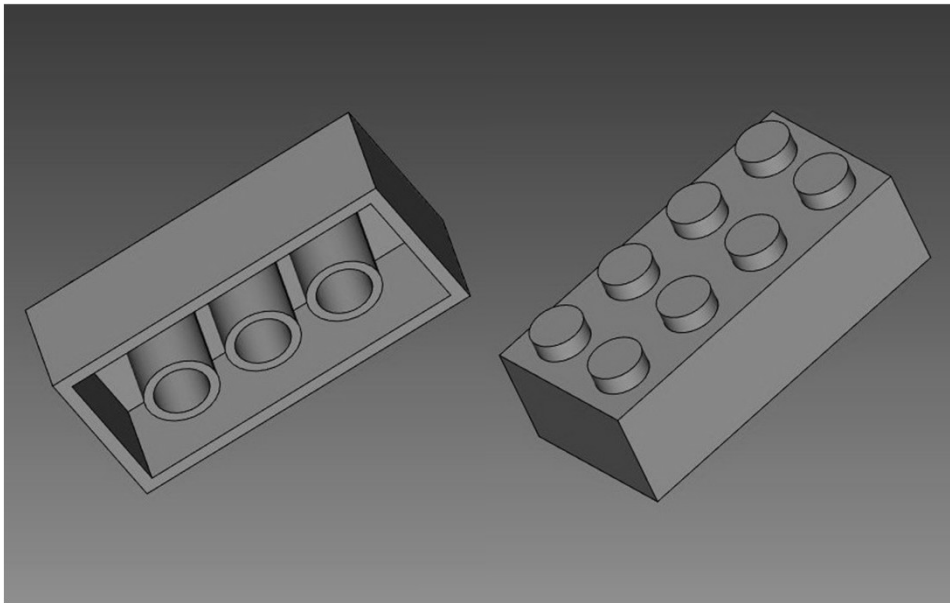


Part Design: use that profile to create a 3D feature

Key rules for beginners

- Green sketch = fully constrained and safer to build on.
- Pad adds material; Pocket removes material.
- Later features should build on earlier ones inside the same Body.
- If the parent sketch changes, the child feature should update with it.
- A clean model tree is more important than fancy shape complexity on day one.

Practice task for the classroom build



Reference tutorial object: base sketch → pad → pocket → top pads

What students will do

1. Create a Body and a first rectangular sketch.
2. Fully constrain the sketch.
3. Pad it into a base solid.
4. Create a second sketch on a face and pocket material out.
5. Add simple circular top features.
6. Change one dimension and confirm the model updates.

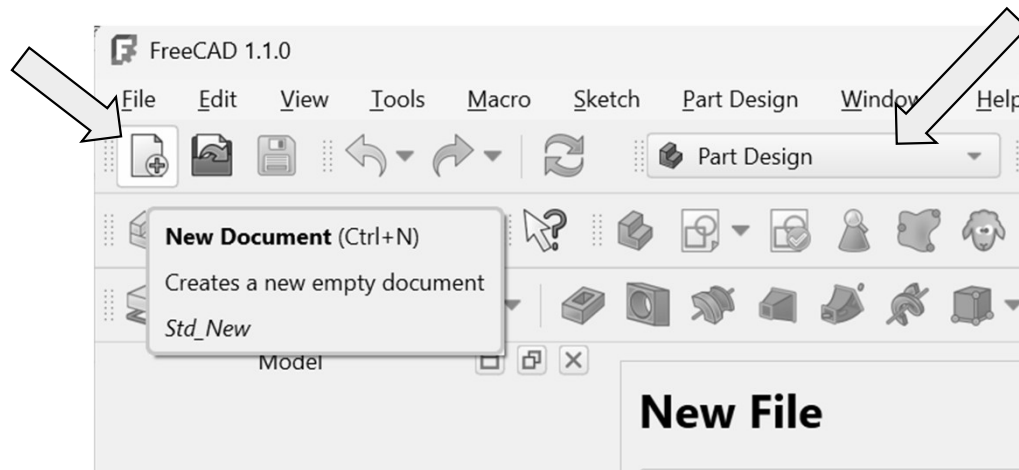
Success criterion: every student leaves class with one editable FCStd file.

- If you finish early: change the base size, pocket depth, or top feature spacing.
- If you fall behind: match the base pad and one pocket first, then continue.

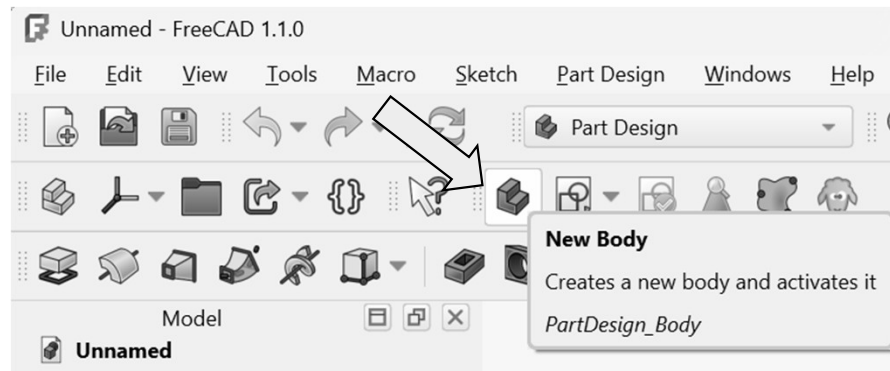
Step 1 — Create a Body and choose the sketch plane

1. Open a new document.

2. Switch to the Part Design workbench.



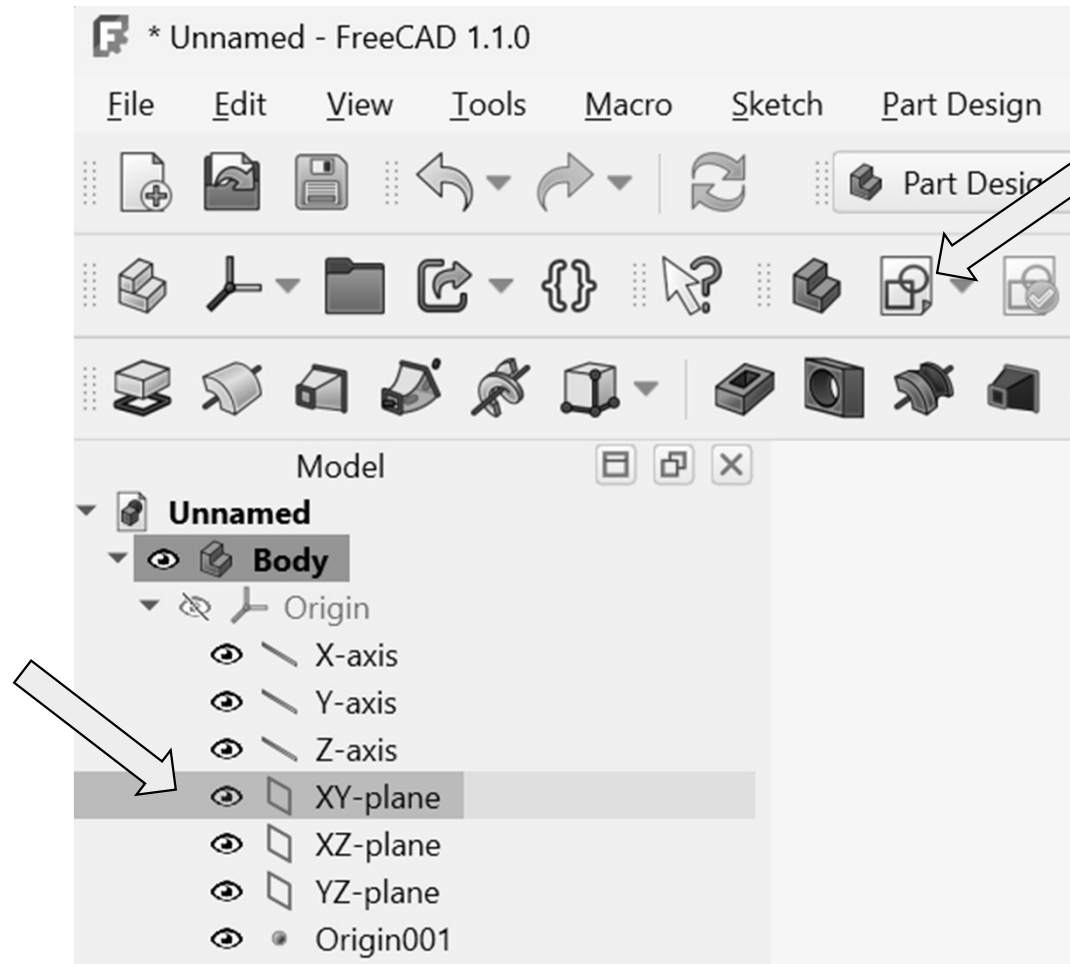
3. Create a Body.



Step 1 — Create a Body and choose the sketch plane

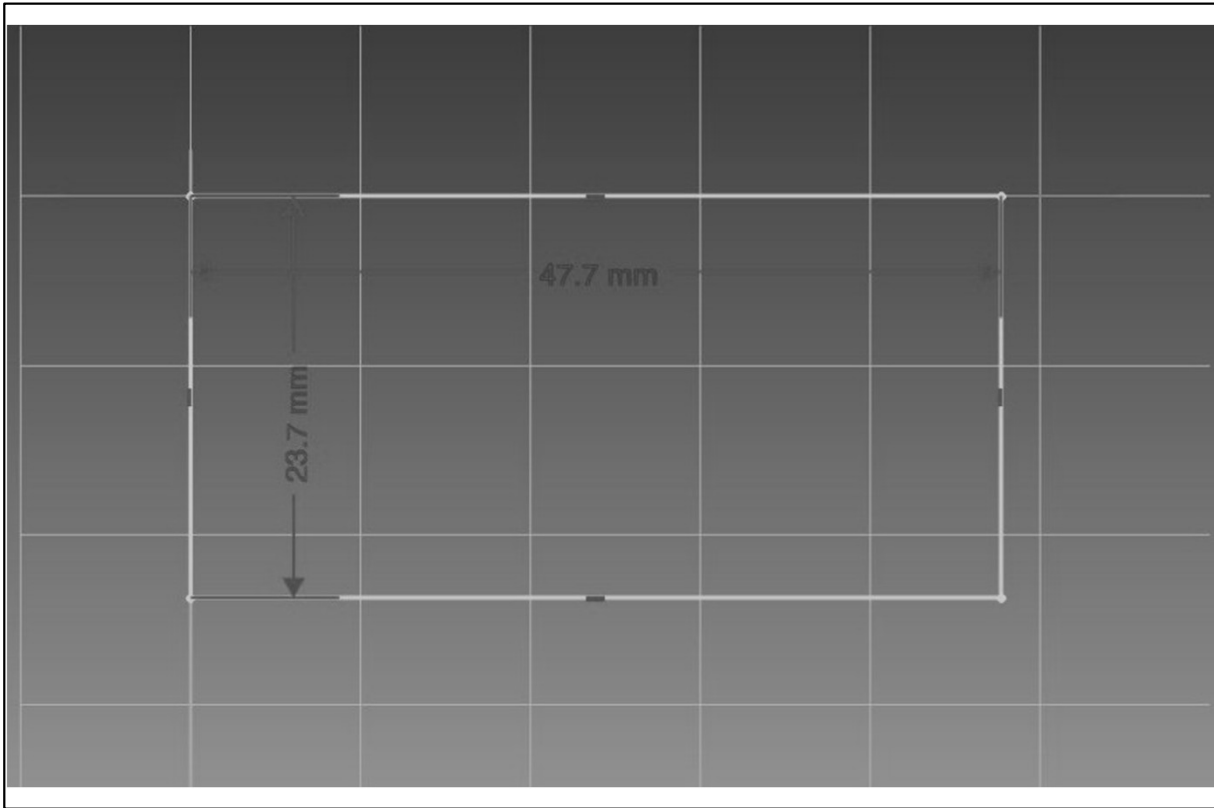
- The plane choice is intentional: it controls the direction of the first pad.

4. Click to choose the base plane.



5. Click Create Sketch

Step 2 — Draw the rectangle and fully constrain it



Tutorial sketch after dimensional and positional constraints

Instructor cues

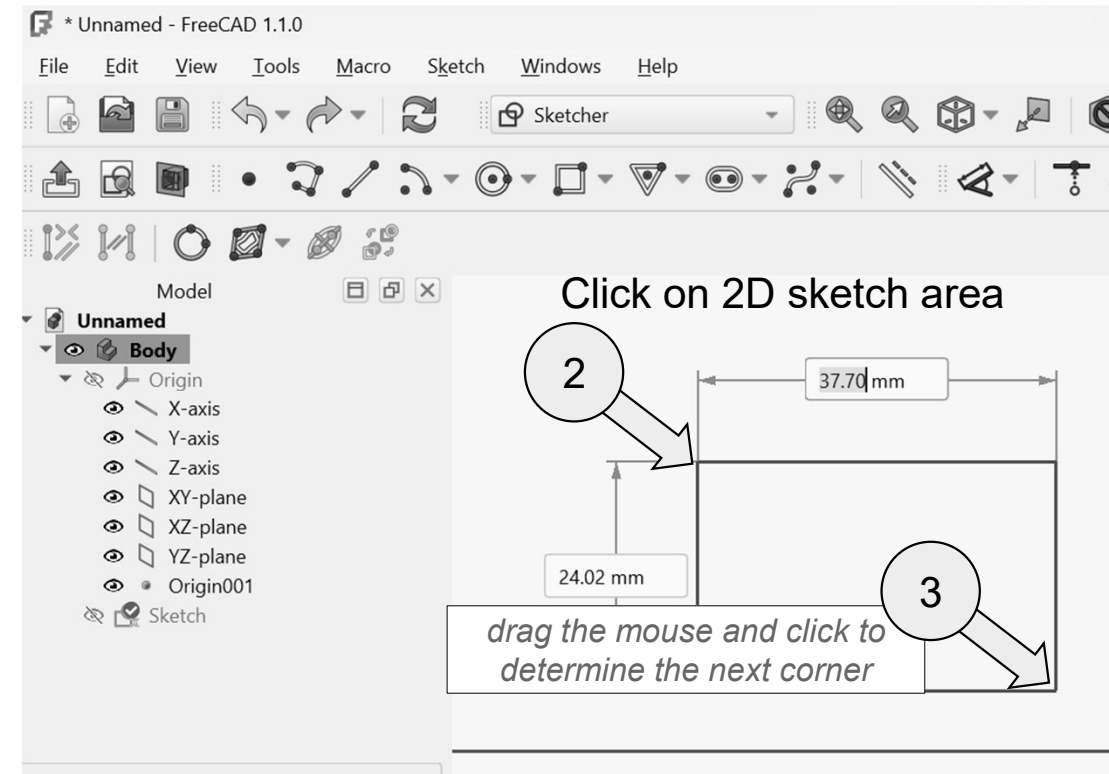
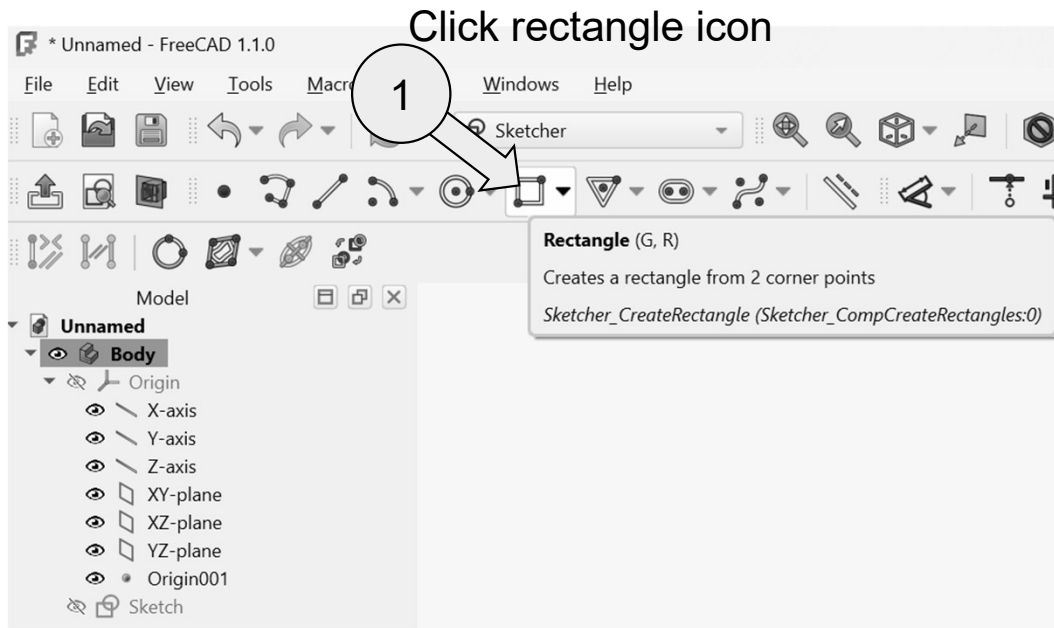
- 1) Draw the rectangle roughly first.
- 2) Then add size constraints.
- 3) Finally anchor the sketch to the origin.
- 4) Green sketch = fully constrained.

Classroom checkpoint

- Ask students to drag a corner lightly.
- If anything still moves, the sketch is not ready for Pad.

Step 2 — Draw the rectangle and fully constrain it

1) Draw the rectangle roughly.



Step 2 — Draw the rectangle and fully constrain it

2) Then add size constraints.

The screenshot shows the FreeCAD 1.1.0 interface with a sketch of a rectangle. The process of adding a dimension is illustrated with numbered steps:

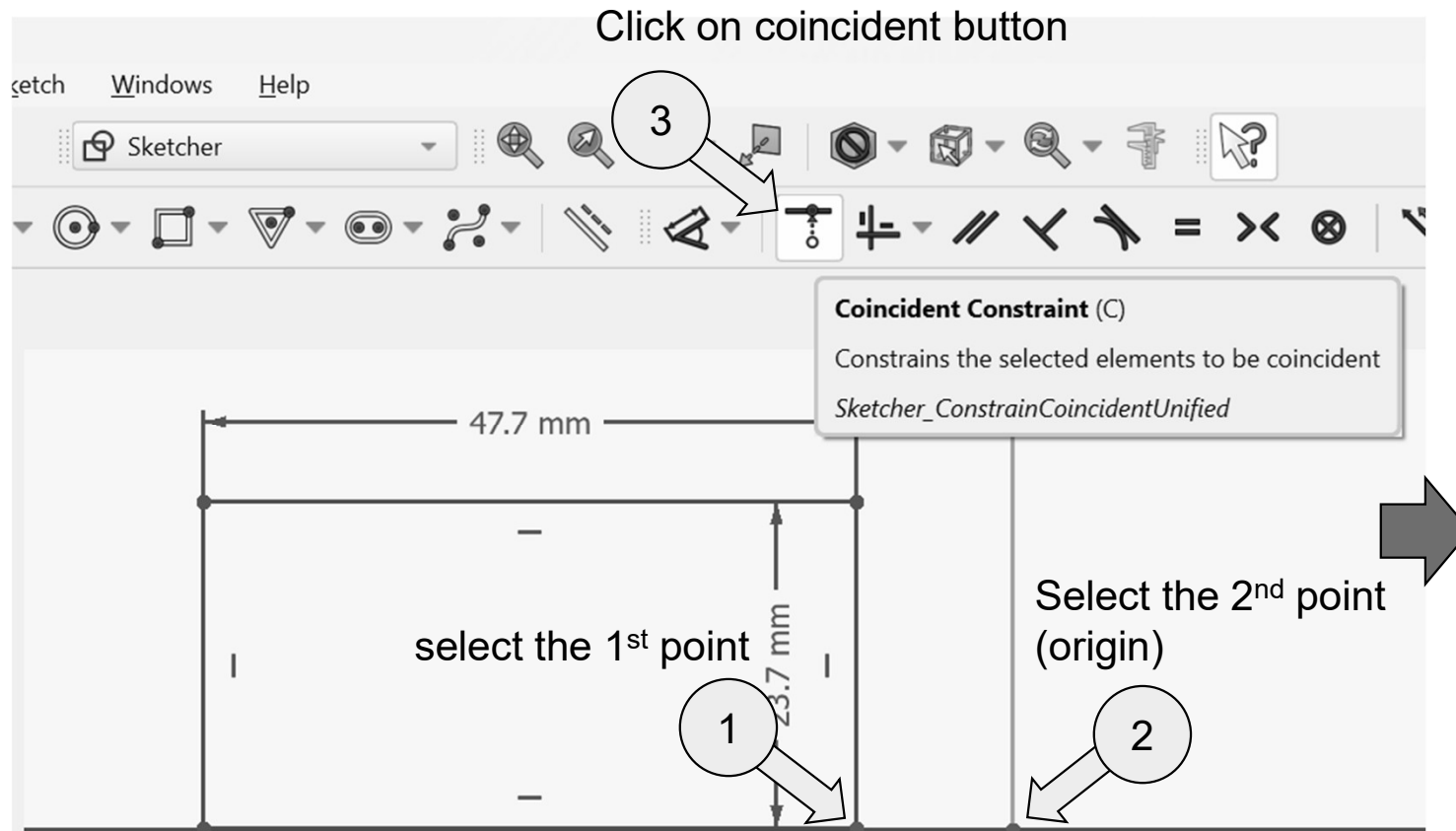
- 1** select the edge: An arrow points to the top edge of the rectangle.
- 2** Click on dimension button: An arrow points to the dimension icon in the top toolbar.
- 3** Click on Horizontal dimension button: An arrow points to the 'Horizontal Dimension' option in the dropdown menu.
- 4** Insert dimension: An arrow points to the 'Insert Length' dialog box, which shows a length of 47.70 mm.
- 5** Click OK: An arrow points to the 'OK' button in the dialog box.

The 'Insert Length' dialog box contains the following fields and buttons:

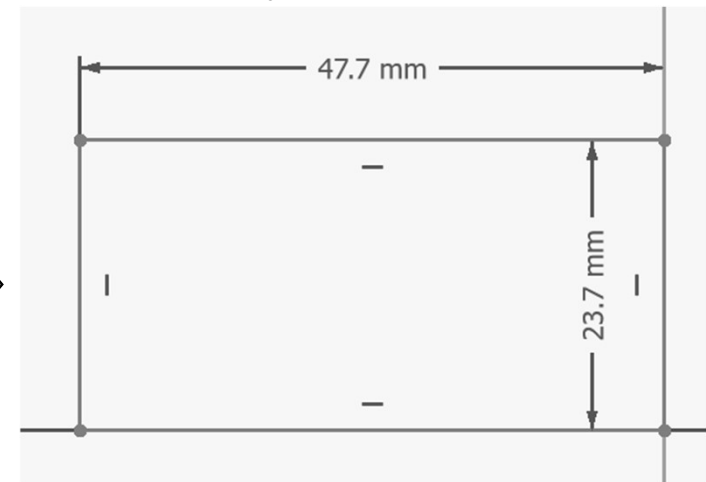
- Length: 47.70 mm
- Name: (empty field)
- ☐ Reference
- OK
- Cancel

Step 2 — Draw the rectangle and fully constrain it

3) Finally anchor the sketch to the origin.



Green = fully constraints



Step 3 — Pad the sketch into the first solid

Pad adds material and stays linked to its sketch, so later sketch edits can recompute the solid.

Leave sketch

1

Click pad

2

Click OK/Enter

3

Set height

3

The image consists of three screenshots from the FreeCAD 1.1.0 interface, illustrating the steps to create a solid from a sketch.

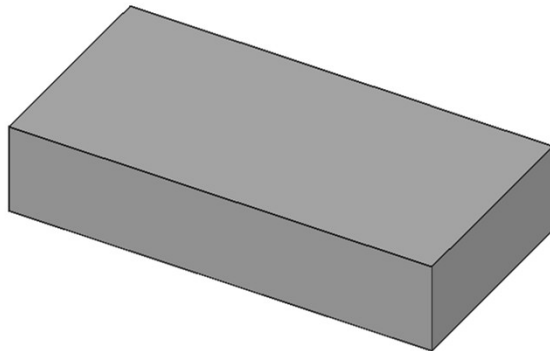
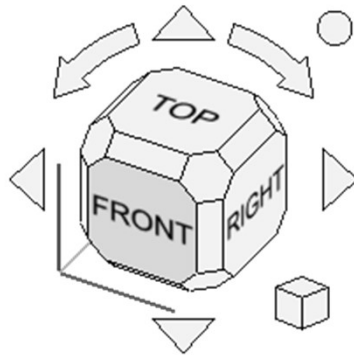
Step 1: Leave sketch
The first screenshot shows the Sketcher toolbar. A callout box labeled '1' points to the 'Leave Sketch' button (represented by an upward arrow). The callout text reads: 'Leave Sketch', 'Exits the active sketch', and 'Sketcher_LeaveSketch'. The left sidebar shows the 'Body' tree with 'Origin' and 'Sketch' listed.

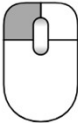
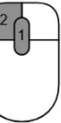
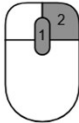
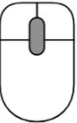
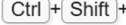
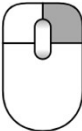
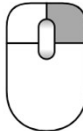
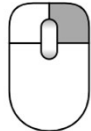
Step 2: Click pad
The second screenshot shows the Part Design toolbar. A callout box labeled '2' points to the 'Pad' button (represented by a block with a downward arrow). The callout text reads: 'Click pad'. The left sidebar shows the 'PartDesign_Pad' feature created under the 'Body'.

Step 3: Click OK/Enter
The third screenshot shows the 'Pad Parameters' dialog box. A callout box labeled '3' points to the 'OK' button. The callout text reads: 'Click OK/Enter'. The dialog box shows the 'Length' field set to '10.00 mm'. Another callout box labeled '3' points to the 'Length' field, with the text 'Set height' above it.

Mouse Navigation

Choosing view

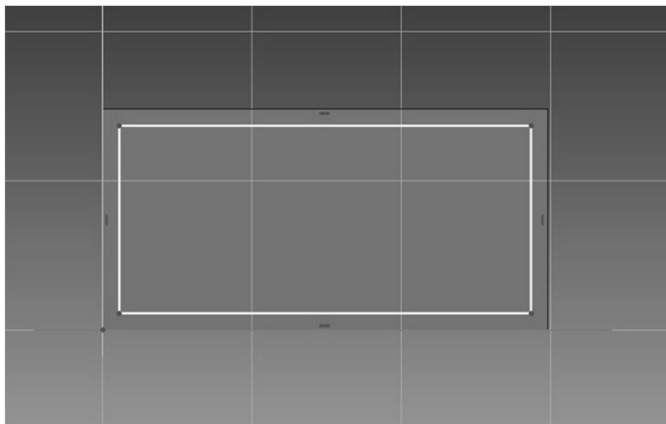


Select	Zoom	Rotate view First method	Rotate view Alternate method	Pan
  Press the left mouse button over an object you want to select.	  Use the mouse wheel to zoom in and out. Clicking the middle mouse button re-centers the view on the location of the pointer.	 Hold the middle mouse button, then press and hold the left mouse button, then move the pointer. If the buttons are released before you stop the mouse motion, the view continues spinning, if this is enabled.	 Hold the middle mouse button, then press and hold the right mouse button, then move the pointer. With this method the middle mouse button may be released after the right mouse button is held pressed. Users who use the mouse with their right hand may find this method easier than the first method.	 Hold the middle mouse button, then move the pointer.
	  Zoom mode: hold the Ctrl and Shift keys, press the right mouse button once, then move the pointer.	  Rotate mode: hold the Shift key, press the right mouse button once, then move the pointer.		  Pan mode: hold the Ctrl key, press the right mouse button once, then move the pointer.

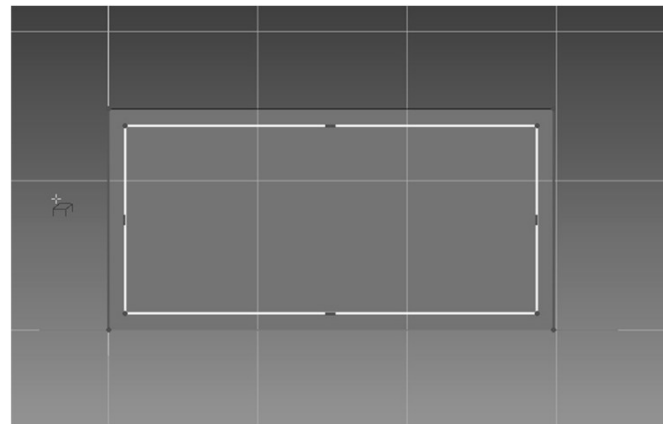
Step 4 — Sketch on a face and use external geometry

- 1) Select the target face before creating the next sketch.
- 2) Draw the internal rectangle on that face.
- 3) Import face edges as reference geometry, then add offsets.

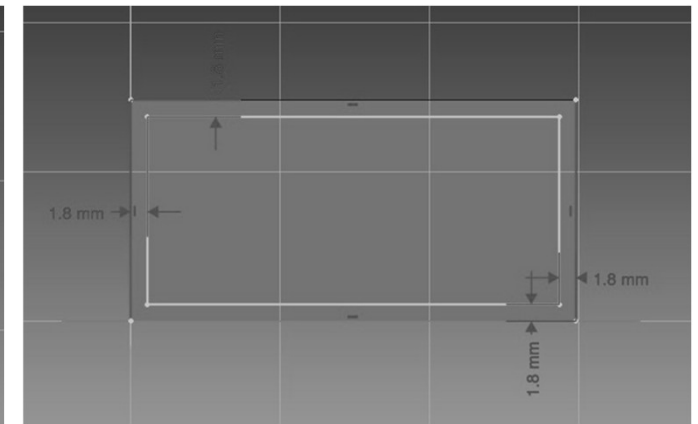
External projection to
import face edges



New sketch on the selected face

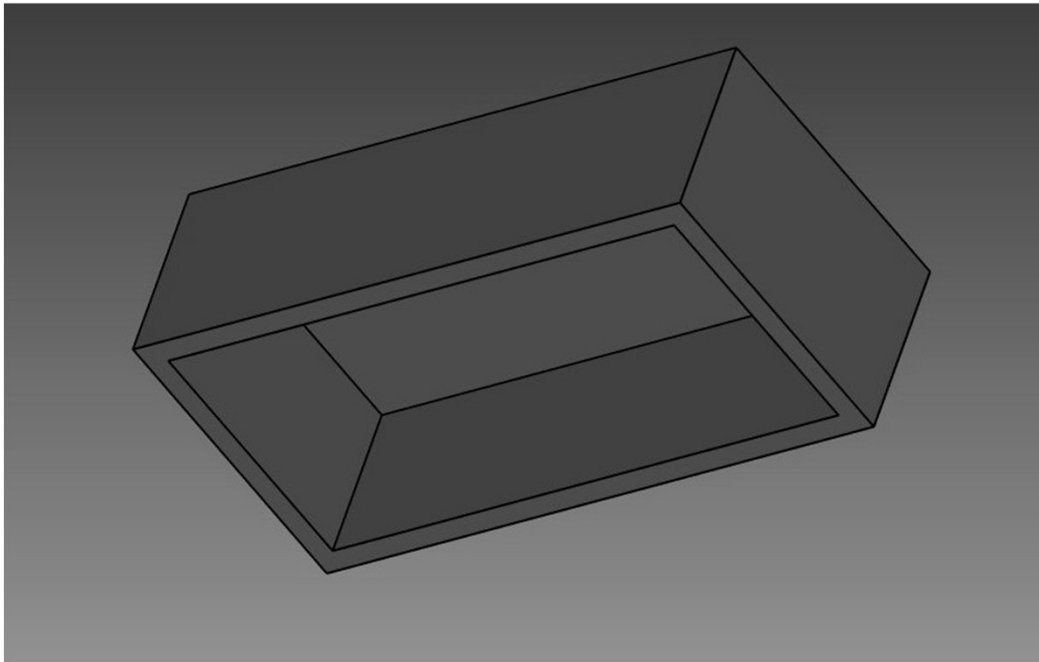


Imported external edges in magenta



Offsets define the pocket boundary

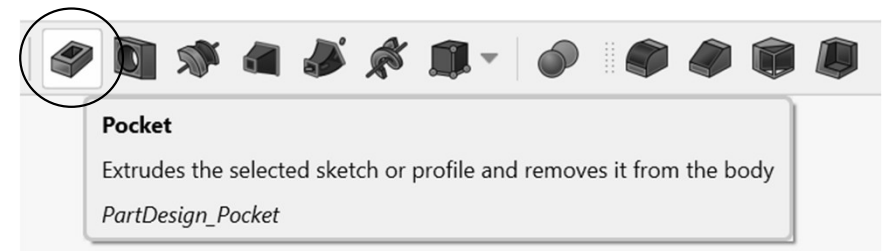
Step 5 — Pocket the interior



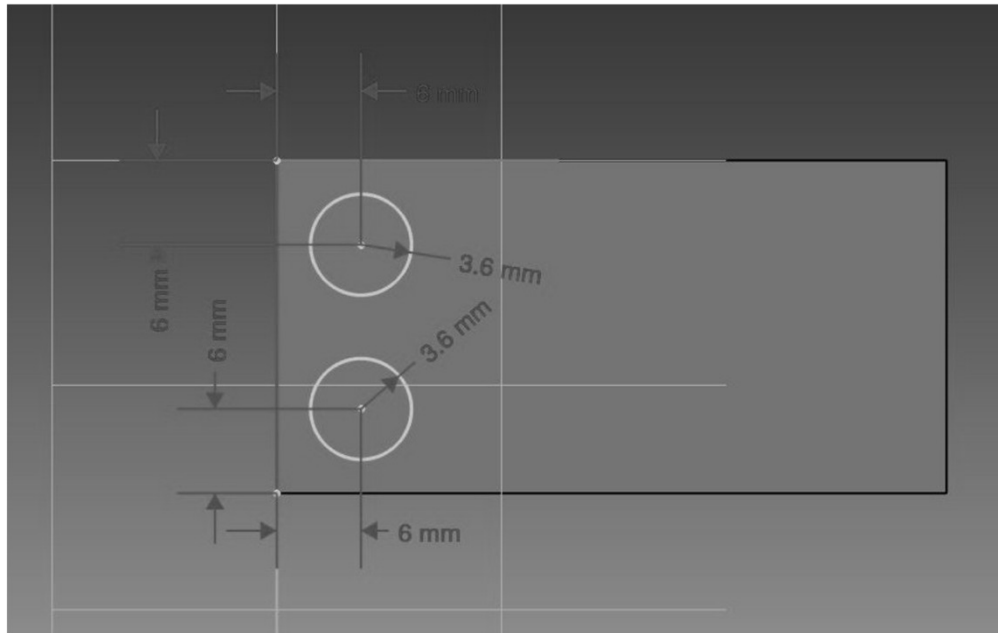
Pocket result after subtracting material

Pocket checklist

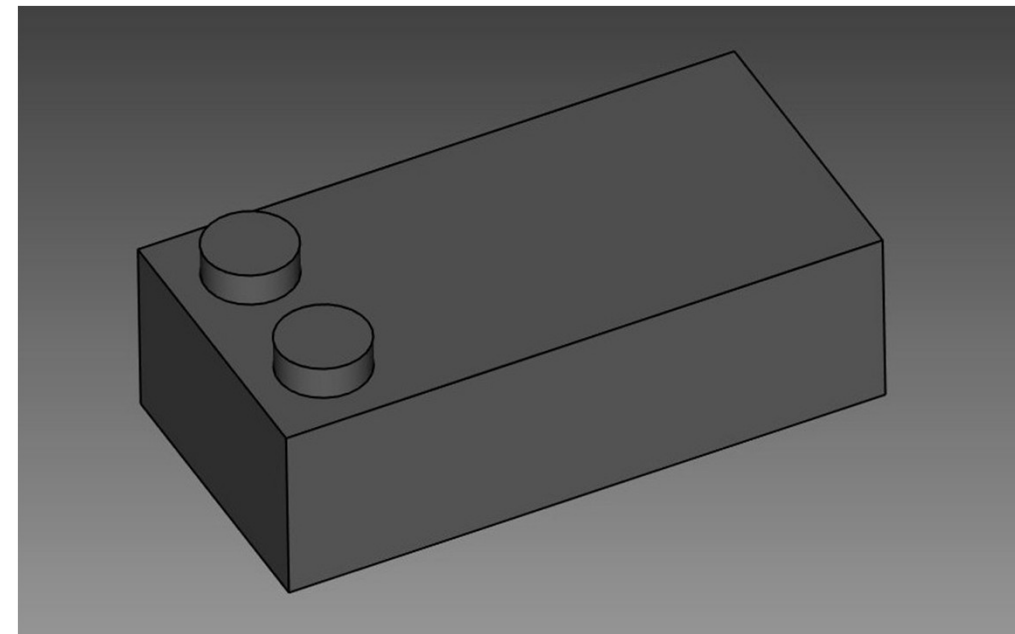
- Leave the sketch edit mode.
- Select the new sketch in the model tree.
- Choose Pocket instead of Pad.
- Set the cut depth or through-all behavior.
- Inspect the cavity from an angled view.



Step 6 — Add circles on the top face, then Pad again



Top-face sketch: circles are dimensioned from imported edges

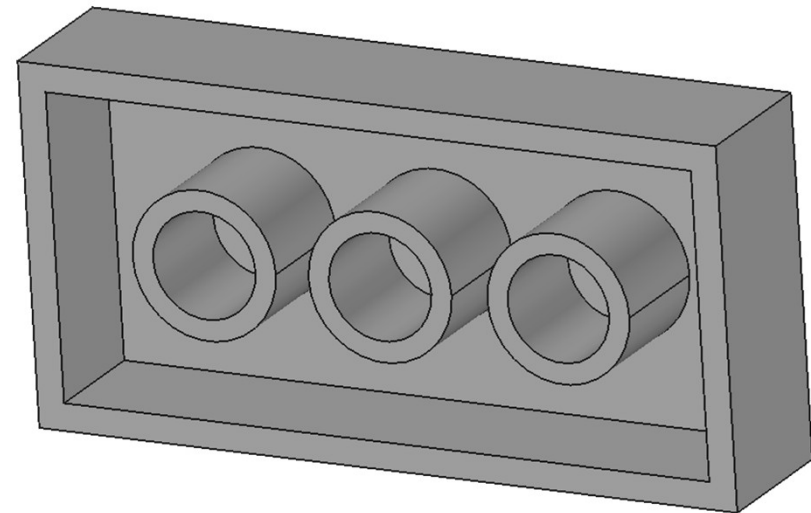
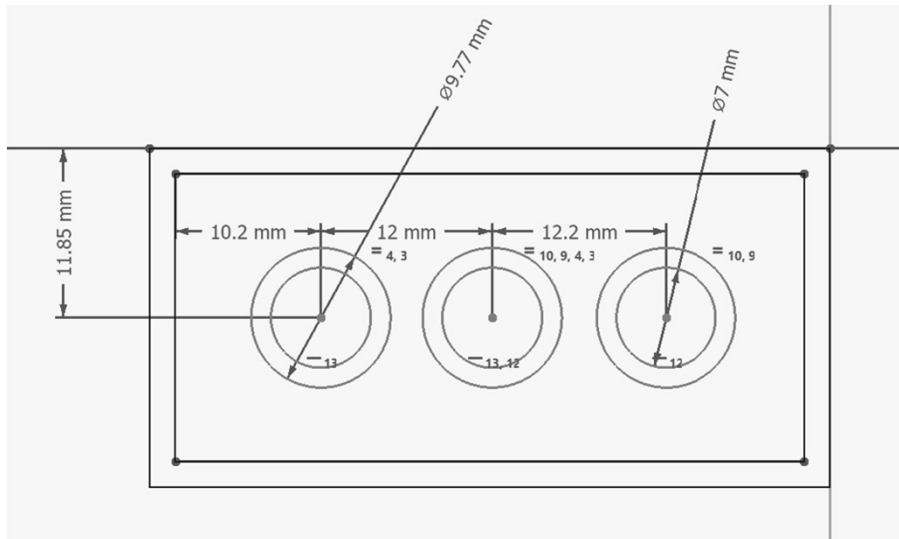


Result after padding the circular sketch

Pattern to remember: select face → sketch → constrain → Pad.

Step 7 — Add circles on the bot face, then Pad again

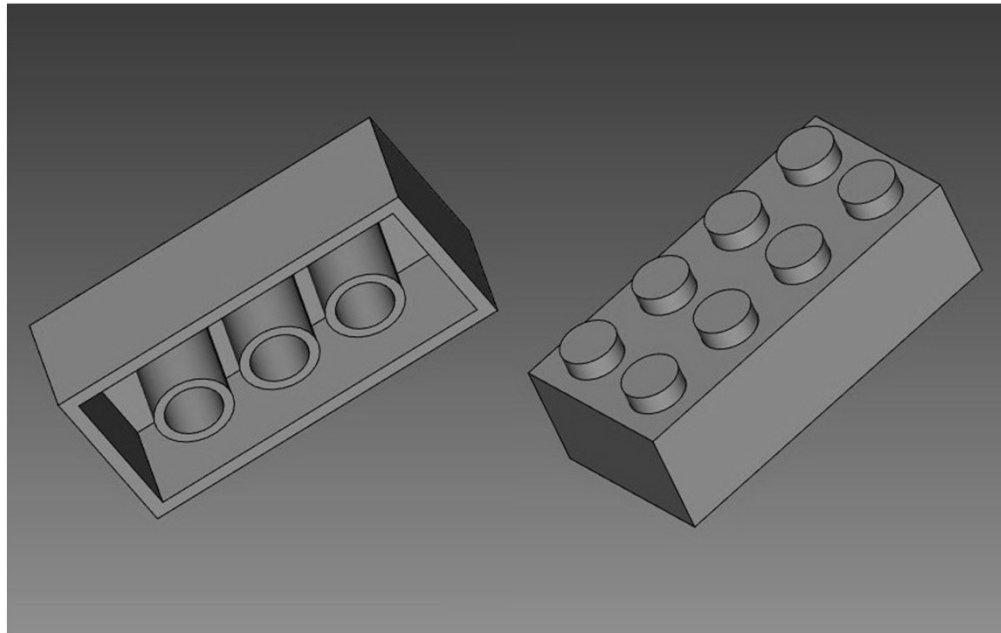
Do the same with step 6, but the circles is at the bot face



Result after padding the circular sketch

Pattern to remember: select face → sketch → constrain → Pad.

Step 8 — Edit a value and test the model



Completed tutorial part: a useful checkpoint before independent edits

In-class parameter edits

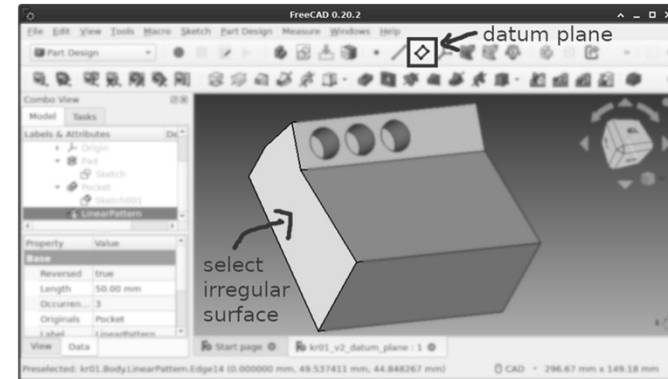
1. Double-click the base sketch and change one dimension.
2. Confirm that the pad updates.
3. Edit pocket depth or top-circle size.
4. Observe how later features follow earlier edits.

Remarks!

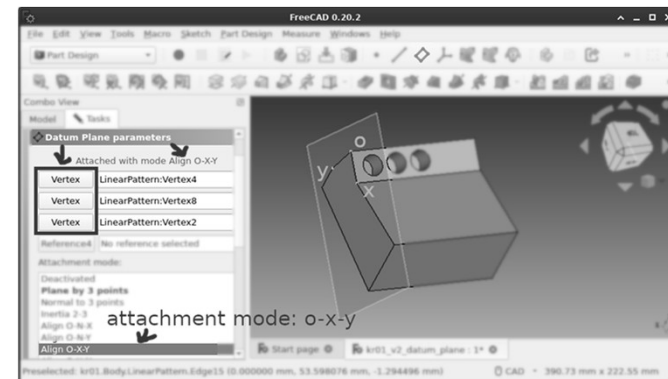
- Parametric modeling is valuable because revision is built in.
- A correct workflow makes change cheaper than rebuilding.

Troubleshooting for beginners

- Sketch still moves → it is under-constrained; do not build on it yet.
- Feature appears in the wrong place → the wrong face or plane was selected before sketching.
- Hole or pocket stops staying centered after edits → the sketch was constrained to the wrong reference.
- A difficult slanted or irregular face may need a datum plane instead of direct face sketching.

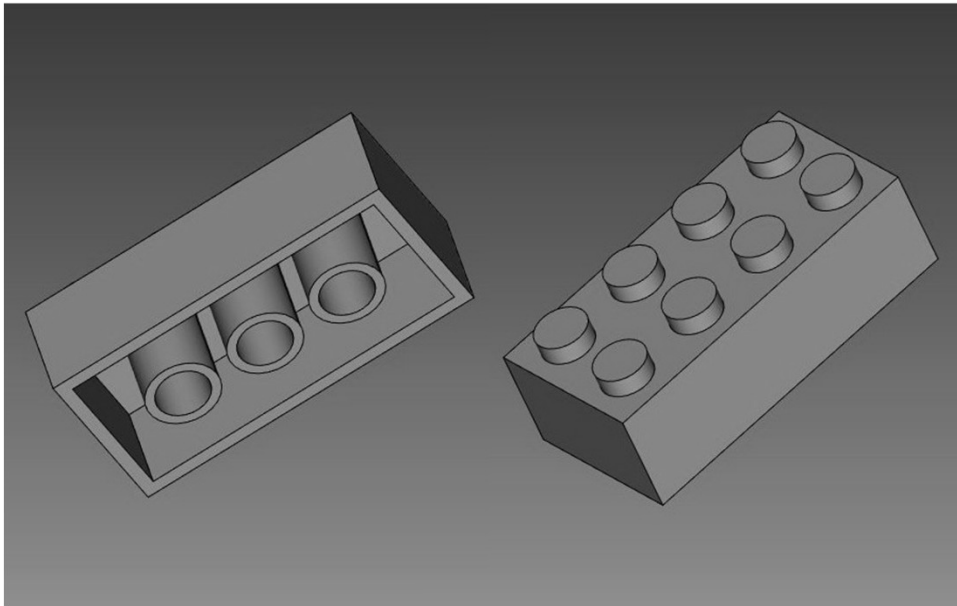


If a face is unsuitable, create a datum plane



Attach the plane carefully so the next sketch has a stable reference frame

Mini challenge



Students modify their own copy rather than starting from zero

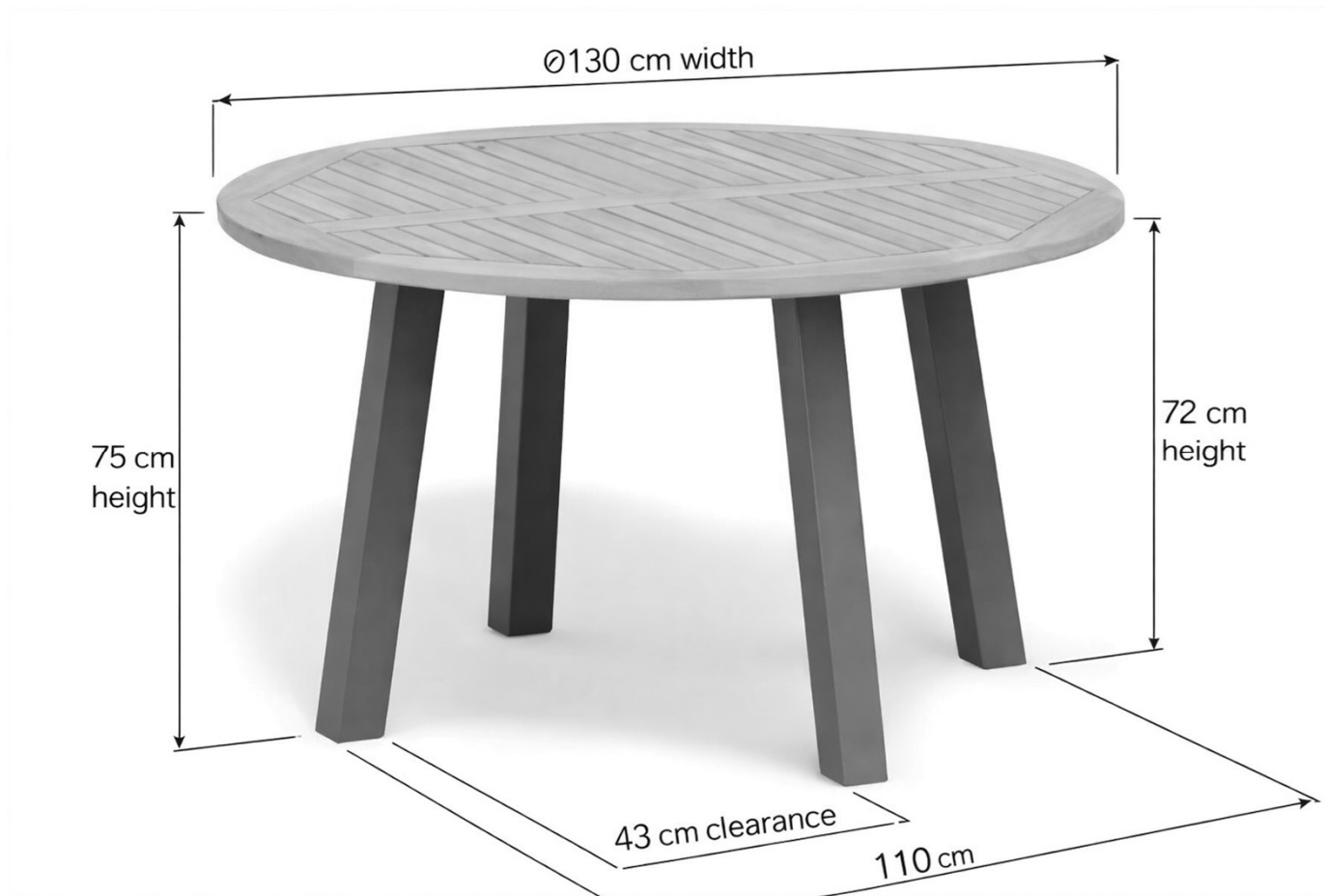
Choose one modification

1. Make the base longer or taller.
2. Change the pocket depth.
3. Resize the circular top features.
4. Add one extra repeated feature if you finish early.

Quick assessment rubric

- The file opens and the Body is valid.
- At least one edited parameter updates the model.
- The student can explain which sketch drives which feature.

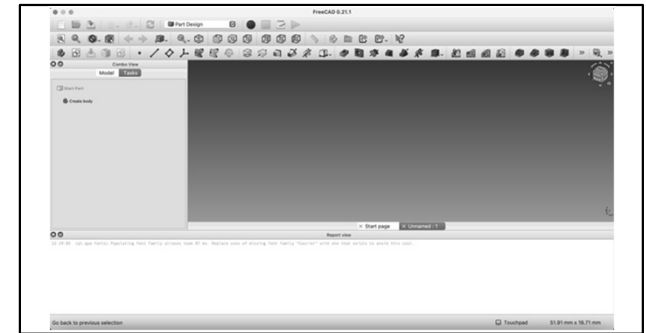
Still have time? Draw this



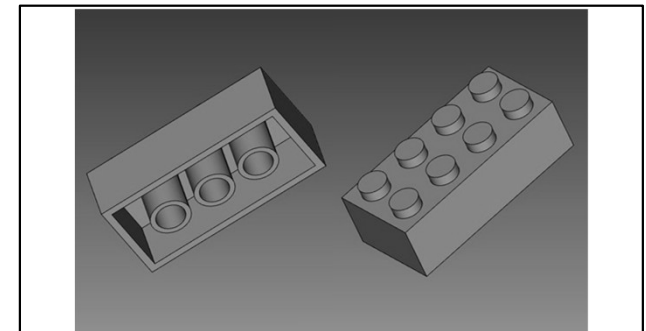
Takeaways

By the end of Lecture 7, students should be able to say:

- A CAD model stores editable design logic, not only what is currently rendered.
- FreeCAD beginner workflow = Body → Sketch → Constraints → Pad / Pocket → Edit values.
- A fully constrained sketch is the safest starting point for later features.
- Simple parts become powerful when they remain easy to revise.



Start page / new document view



A finished but still editable result